

DSA0601 - DATA HANDLING AND VISUALISATION

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Question 1 – Empirical Cumulative Distribution Function (ECDF)

Scenario

A teacher recorded the marks of students in a Data Science quiz. The teacher wants to

determine the cumulative proportion of students scoring at or below a given mark.

Dataset

Student Marks

S1 45

S2 52

S3 58

S4 60

S5 65

S6 68

S7 72

S8 75

S9 80

S10 90

Question

Construct the Empirical Cumulative Distribution Function (ECDF) for the above dataset.

Calculate the cumulative probability for each observation and interpret the ECDF plot.

CODE:

```
marks <- c(45, 52, 58, 60, 65, 68, 72, 75, 80, 90)
```

```
ecdf_marks <- ecdf(marks)
```

```
data.frame(
```

```

Marks = sort(marks),

Cumulative_Probability = ecdf_marks(sort(marks))

)

plot(ecdf_marks,

    main = "ECDF of Student Marks",

    xlab = "Marks",

    ylab = "Cumulative Probability",

    col = "blue",

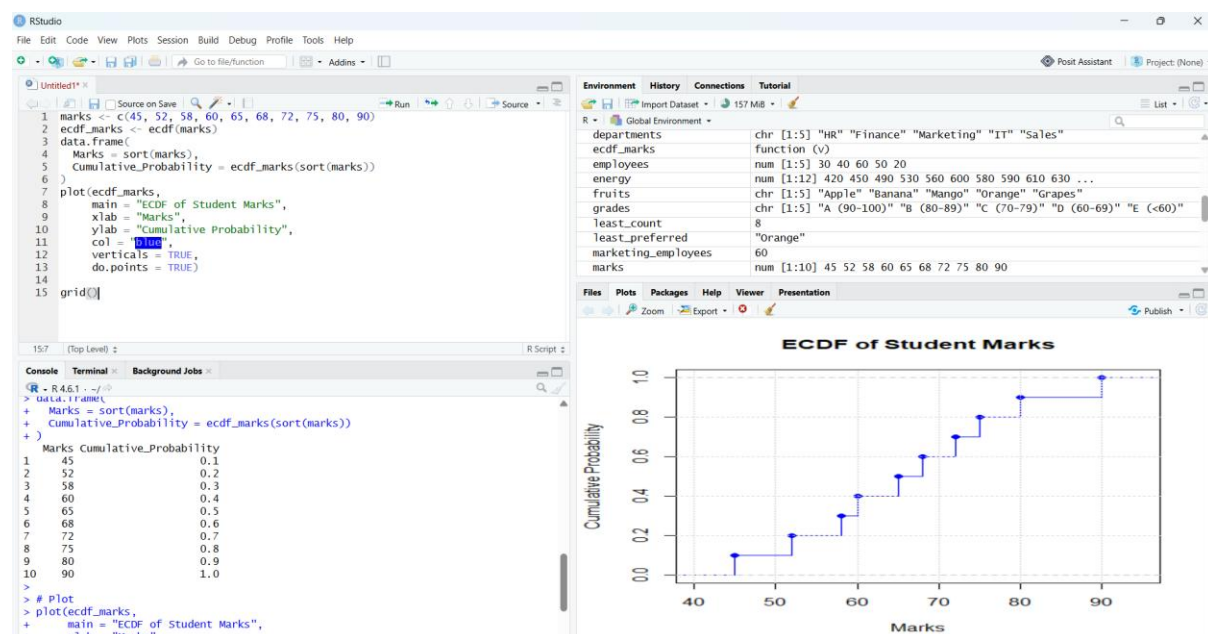
    verticals = TRUE,

    do.points = TRUE)

grid()

```

Output:



Question 2 – Empirical Cumulative Distribution Function (ECDF)

Scenario

A fitness trainer recorded the number of daily steps completed by ten participants during a

health challenge.

Dataset

Participan Daily

t Steps

P1 4200

P2 5000

P3 5600

P4 6100

P5 6800

P6 7200

P7 8100

P8 9200

P9 10400

P10 12000

Question

Using the dataset, create an ECDF plot. Determine the proportion of participants who

walked 8,000 steps or fewer and explain the significance of the ECDF.

CODE:

```
steps <- c(4200, 5000, 5600, 6100, 6800,
```

```
          7200, 8100, 9200, 10400, 12000)
```

```
ecdf_steps <- ecdf(steps)
```

```
prop <- ecdf_steps(8000)
```

```
print(prop)
```

```
plot(ecdf_steps,
```

```
     main = "ECDF of Daily Steps",
```

```
     xlab = "Daily Steps",
```

```
     ylab = "Cumulative Probability",
```

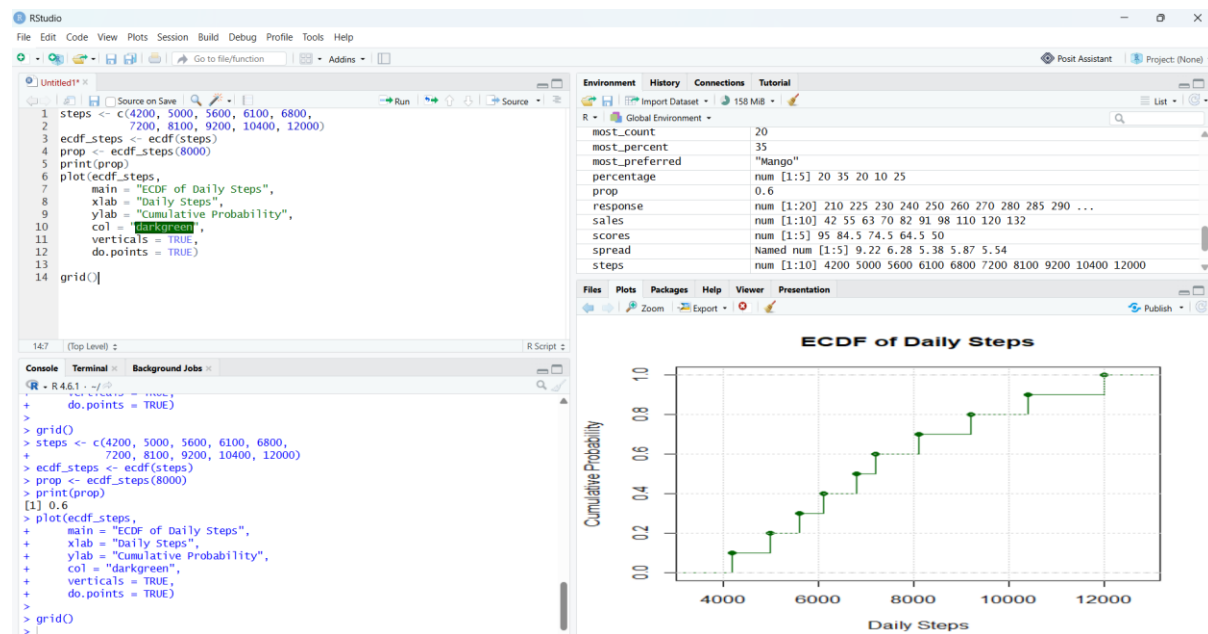
```
col = "darkgreen",
```

```
verticals = TRUE,
```

```
do.points = TRUE)
```

```
grid()
```

OUTPUT:



Question 3 – Highly Skewed Distribution (Histogram and Density Plot)

Scenario

An online shopping company recorded the purchase amount (₹) made by customers during

a promotional sale. Most customers purchased inexpensive items, while a few made very

expensive purchases.

Dataset Customer Purchase Amount (₹)

C1 250

C2 300

C3 320

C4 350

C5 380

C6 450

C7 520

C8 650

C9 1800

C10 5200

Question

Create a Histogram and a Density Plot for the purchase amounts. Determine whether the distribution is positively skewed or negatively skewed, and justify your answer.

CODE:

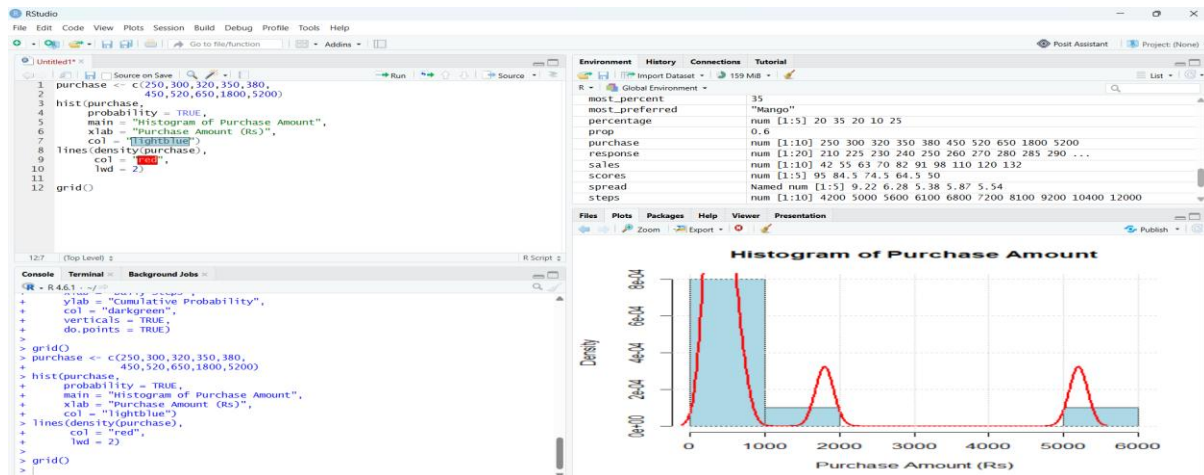
```
purchase <- c(250,300,320,350,380,
              450,520,650,1800,5200)

hist(purchase,
     probability = TRUE,
     main = "Histogram of Purchase Amount",
     xlab = "Purchase Amount (Rs)",
     col = "lightblue")

lines(density(purchase),
     col = "red",
     lwd = 2)

grid()
```

OUTPUT:



Question 4 – Quantile-Quantile (Q-Q) Plot

Scenario

A hospital administrator collected the waiting times (in minutes) for patients in the outpatient department and wants to verify whether the data follows a normal distribution.

Dataset

Patient Waiting Time (minutes)

P1 12

P2 15

P3 18

P4 20

P5 22

P6 24

P7 25

P8 28

P9 30

P10 32

Question

Construct a Normal Q-Q Plot using the waiting time data. Based on the plot, determine

whether the waiting times approximately follow a normal distribution.

CODE:

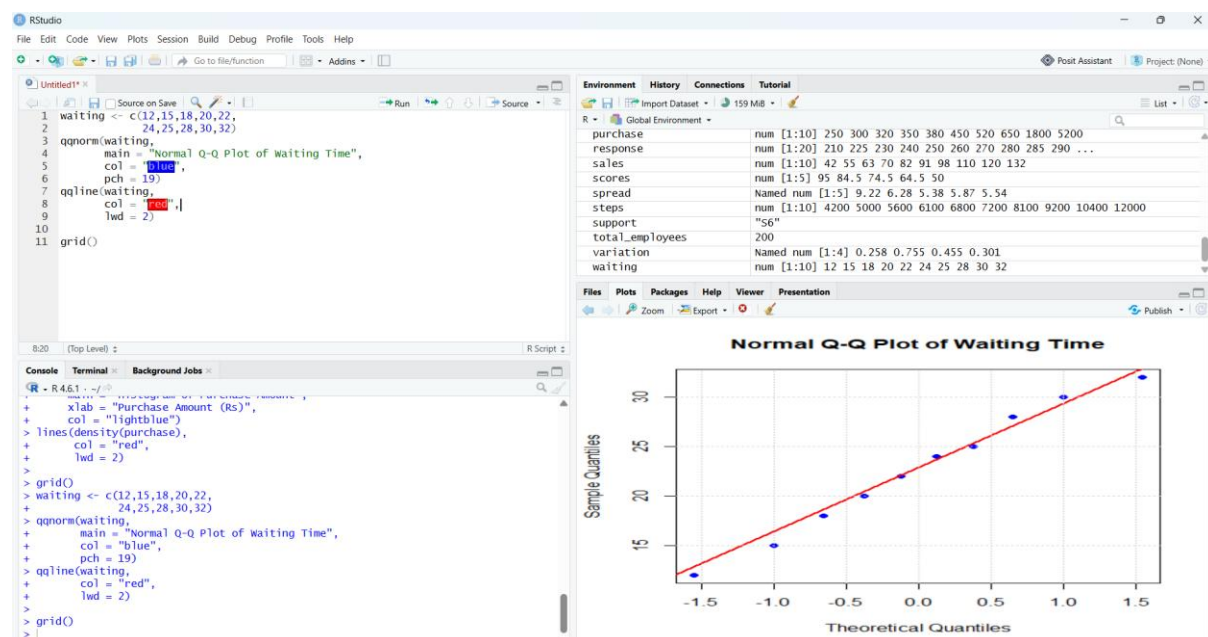
```
waiting <- c(12,15,18,20,22,
            24,25,28,30,32)

qqnorm(waiting,
        main = "Normal Q-Q Plot of Waiting Time",
        col = "blue",
        pch = 19)

qqline(waiting,
        col = "red",
        lwd = 2)

grid()
```

OUTPUT:



Question 5 – Highly Skewed Distribution and Q-Q Plot

Scenario

A bank analyzed the monthly income (₹) of ten customers applying for home loans.

A few

customers have exceptionally high incomes compared to the majority.

Dataset

Customer Monthly Income (₹)

C1 18000

C2 22000

C3 24000

C4 26000

C5 28000

C6 30000

C7 34000

C8 42000

C9 85000

C10 200000

CODE:

```
income <- c(18000,22000,24000,26000,28000,
            30000,34000,42000,85000,200000)

hist(income,
     main = "Histogram of Monthly Income",
     xlab = "Monthly Income (Rs)",
     col = "lightgreen")

grid()

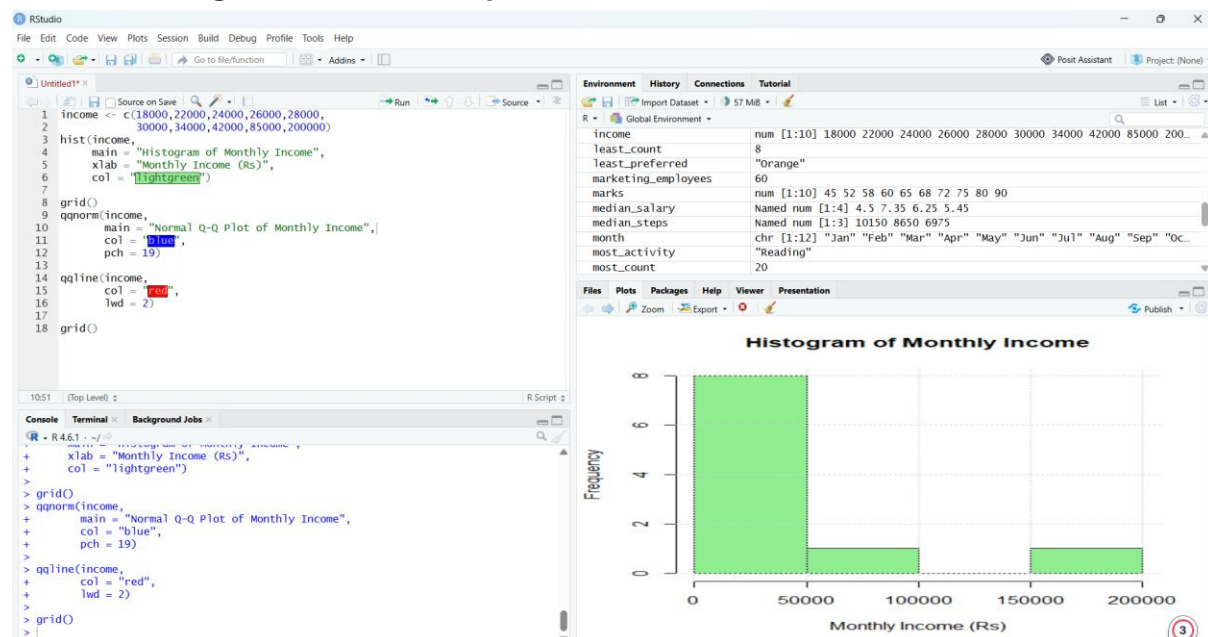
qqnorm(income,
      main = "Normal Q-Q Plot of Monthly Income",
      col = "blue",
      pch = 19)

qqline(income,
      col = "red",
      lwd = 2)
```

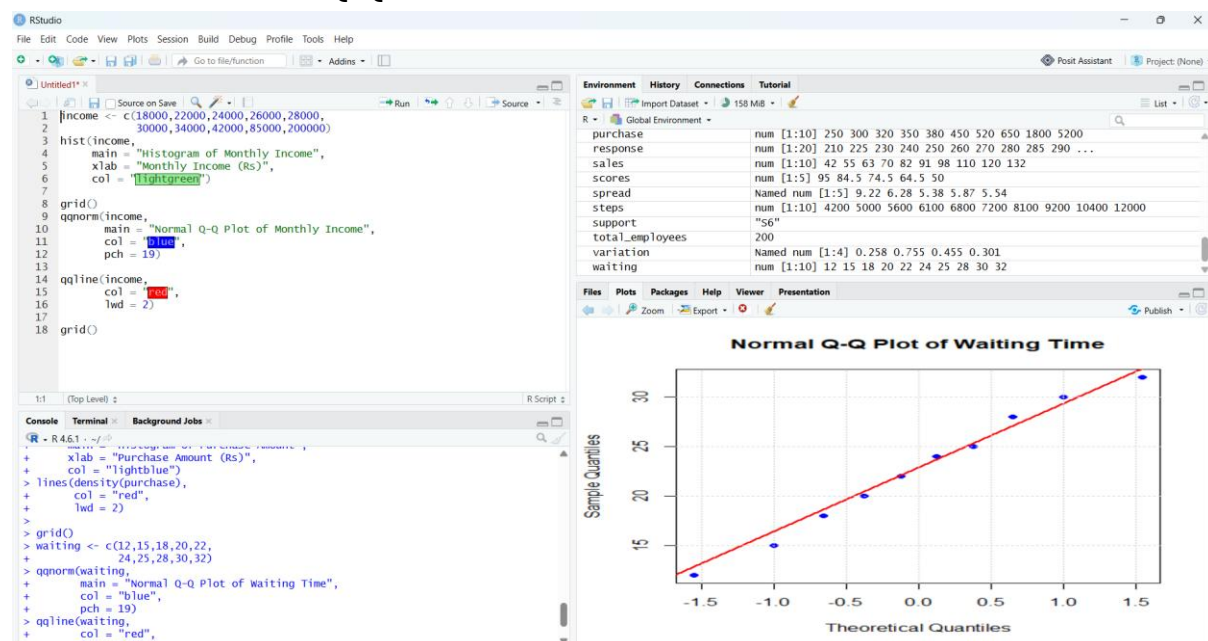
grid()

Question

1. Plot a Histogram of the monthly income data.



2. Generate a Normal Q-Q Plot.



3. Determine whether the income data is normally distributed or highly skewed.

The monthly income data is highly positively (right) skewed and not normally distributed. Most customers have incomes between ₹18,000 and ₹42,000, while two customers have much higher incomes (₹85,000 and ₹200,000). These extreme values create a long right tail in the distribution.

4. Explain how the Q-Q plot supports your conclusion.

The Normal Q-Q Plot shows that the data points do not lie close to the reference straight line, especially at the upper end. The points corresponding to the highest incomes (₹85,000 and ₹200,000) deviate significantly from the line, indicating the presence of outliers and a strong right skew. Therefore, the Q-Q plot confirms that the income data does not follow a normal distribution and is highly positively skewed.